INTERNATIONAL MONETARY FUND

REPUBLIC OF TAJIKISTAN

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REQUEST FOR DISBURSEMENT UNDER THE RAPID CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS

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Tajikistan: Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*High*
Overall risk of debt distress	*High*
Granularity in the risk rating	*Sustainable*
Application of judgment	*No*

This joint World Bank/ IMF Debt Sustainability Analysis (DSA) [1] *indicates that Tajikistan's overall risk of debt distress remains high (unchanged from the December 2019 DSA). Public debt is expected to spike above 50 percent of GDP in 2020 in reaction to the outbreak of the COVID-19 pandemic. However, based on the authorities' commitment to fiscal consolidation, debt stabilizes over the medium term and is assessed as sustainable in a forward-looking sense (changed from the December 2019 DSA where debt was on a rising path).*

Under the baseline, two debt indicators breach thresholds, leading to a high external risk rating. With planned medium-term fiscal consolidation program (including adjustment of 2 percent of GDP in 2021-2022) and the commitment to avoid non-concessional borrowing, these indicators are projected to stabilize below or near respective thresholds by 2030.

Tajikistan's debt is vulnerable, especially to export shocks and contingent fiscal liabilities. A more severe or prolonged COVID-19 shock could heighten vulnerabilities. Medium- and long-term fiscal consolidation, avoiding non-concessional external borrowing, diversifying exports, and containing contingent liabilities from SOEs would help reduce vulnerabilities and stabilize debt.

[1] Tajikistan is assessed to have a medium debt-carrying capacity. CI rating for Tajikistan is 2.90, which is based on the October 2019 WEO and the World Bank's 2018 CPIA.

COVERAGE AND BACKGROUND ON PUBLIC DEBT

A. Background on Debt and Debt Coverage

1. In recent years, external and financial sector vulnerabilities have contributed to an increase in debt. Tajikistan's external public- and publicly-guaranteed (PPG) debt rose from 24 percent of GDP in 2014 to near 36 percent of GDP at end-2019 mainly as a consequence of a sizable depreciation of the somoni as well as the issuance of a $500 million sovereign bond in 2017. Domestic PPG debt also increased from 3½ percent of GDP at end-2014 to 8 percent of GDP at end-2019, owing to a 6 percent of GDP recapitalization of banks in December 2016. As a result, the total PPG debt increased from 27½ percent of GDP in 2014 to 44.6 percent of GDP in 2019.

2. External debt made up the bulk of the total of PPG debt in 2019. External PPG debt accounted for about 81 percent of total PPG debt. Over 80 percent of external PPG debt was owed to multilateral and bilateral creditors. The single largest creditor was China, which held over 40 percent of the total PPG external debt. The Eurobond of USD 500 million issued in September 2017, with a maturity of 10 years, carried an interest rate of 7.125 percent.

3. The NBT is the main creditor and holder of largely non-marketable domestic government debt. Most of the government securities held by the NBT were issued at significantly below market terms, with some interest rates as low as 0.99 percent. Since 2016, the government has been accumulating interest and principal arrears to the NBT. In 2019, the arrears on domestic government securities issued for the NBT recapitalization were cleared after the NBT extended new credit to the government at a 2 percent interest rate with a one-year maturity. However, the government continues to run arrears against the NBT on bonds issued to recapitalize commercial banks during the 2015-16 shocks.

4. This DSA covers the central government, central bank, and government-guaranteed external and domestic debt. Debt coverage includes duly consolidated overall external and domestic debt and guarantees of the Central Government (CG), including extrabudgetary funds, and the social security fund. As debt recording and monitoring capacity is weak, this DSA does not include in its baseline: i) non-guaranteed liabilities of state-owned enterprises (SOEs), including liabilities associated with the modernization of an aluminum plant and the construction of a gas pipeline[2], (ii) contingent liabilities/fiscal costs associated with

[2] The Ministry of Finance does not record non-guaranteed debt of SOEs. In 2019, TALCO, a loss-making state-owned aluminum company, was allowed to borrow without a government guarantee. Subsequently, TALCO has signed a MOU to borrow $545 million from China to modernize its plant. The finalization of contract and associated disbursement are expected in 2020. Separately, Tajiktransgaz has borrowed $300 million from Chinese entities for the construction of the Tajikistan section of the Turkmenistan-China gas pipeline.

potential liquidation of two large and troubled financial institutions or iii) demand or guarantees triggered from any existing PPP agreements.[3]

	Subsectors of the public sector	Check box
1	Central government	X
2	State and local government	
3	Other elements in the general government	X
4	o/w: Social security fund	X
5	o/w: Extra budgetary funds (EBFs)	X
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

BACKGROUND ON MACRO FORECASTS

5. The assumptions in the baseline scenario are consistent with macroeconomic framework presented in the staff report.[4] The main assumptions are:

- ***External.*** The current account deficit is expected to widen in 2020, but narrow over the medium term with fiscal consolidation. Remittances are expected to decline sharply in 2020 owing to a more challenging regional and global environment. Electricity exports are expected to rise in 2021 and over the medium term owing to Roghun. International reserves are supported by domestic purchases of monetary gold despite pressures from the balance of payments.

- ***Interest rates***. Effective average interest rates on external debt are projected to rise gradually over the medium-term as concessional financing is likely to be constrained and the authorities would gradually increase reliance on non-concessional external borrowing (after Roghun power purchase agreements are signed and debt is considered sustainable). Interest rates on domestic public debt, some of which are highly negative in real terms at present, are expected to remain below market rates.[5]

- ***Fiscal***. The fiscal deficit is expected to increase sharply in 2020 owing to lower revenues and higher health and social spending as a result of the COVID-19 pandemic. Over the medium-term, in line with authorities' plans for fiscal consolidation, the fiscal deficit is expected to decline to about 2½ percent of GDP. Spending on Roghun and other large infrastructure projects is expected to be accommodated by cuts to other non-priority spending.

- ***Growth.*** A sizable deceleration in 2020 is expected owing to the COVID19 shock, border closures, and associated drop in remittances. This will be followed by a recovery in 2021. Over the medium

[3] State and local governments are not allowed to borrow without a government guarantee.

[4] The baseline includes one disbursement of 80 percent of quota under the Rapid Credit Facility in 2020.

[5] Staff's baseline scenario assumes that the government continues to rely on external financing over the horizon. Over time, the development of domestic debt markets would be advisable. While such borrowing is likely to be more expensive, it would help reduce the vulnerabilities to external shocks.

term, a weak global environment, and uneven structural reforms are expected to weigh on growth. Inflation is expected to remain moderate.

6. Macroeconomic assumptions under the current baseline scenario project a lower fiscal deficit over the medium term than under the 2019 DSA. While real GDP growth estimates for the current year were larger in the 2019 DSA, medium term growth rate projections are broadly similar. A key difference is that the projected fiscal deficit in 2020 has risen as projections now incorporate a drop in revenues and an increase in government spending on treatment and containment costs for COVID-19. The projections also incorporate the government's commitments on 2 percent of GDP in fiscal consolidation measures in 2021-22, which are to be incorporated in the government's budgets in 2021-22 (see LOI ¶ 6). The authorities plan medium term fiscal consolidation to achieve fiscal deficit targets of 4.4 and 2.6 percent of GDP in 2021 and 2022, respectively, to stabilize debt. The external position was projected to remain weak in the 2019 DSA but is expected to improve because of fiscal consolidation. International reserves (in months of imports) are higher in the current DSA owing to domestic purchases of gold but are expected to decline gradually over the medium-term. The DSA also incorporates debt relief from the IMF's Catastrophe Containment and Relief Trust (USD 11 million) and disbursement under the Rapid Credit Facility (USD 193 million).

Tajikistan: Baseline DSA Assumptions, 2017-24

(In percent of GDP)

	2017	2018	2019	2020	2021	2022	2023	2024
Real GDP growth, percent								
2020 DSA	7.1	7.3	7.5	-2.0	7.5	4.5	4.0	4.0
2019 DSA	7.1	7.3	5.5	4.8	4.5	4.5	4.0	4.0
2017 DSA	4.5	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Overall fiscal balance (incl. PIP)								
2020 DSA	-6.0	-2.8	-2.1	-7.7	-4.4	-2.6	-2.5	-2.5
2019 DSA	-6.0	-2.8	-3.8	-4.3	-4.3	-4.3	-4.3	-4.3
2017 DSA	-6.5	-5.4	-1.9	-1.8	-1.7	-1.7	-2.0	-2.2
Current account balance								
2020 DSA	2.2	-5.0	-2.3	-7.8	-4.7	-4.6	-4.4	-4.3
2019 DSA	2.2	-5.0	-4.5	-5.2	-5.3	-5.3	-5.4	-5.5
2017 DSA	-6.3	-6.2	-5.6	-5.1	-4.6	-4.3	-4.3	-4.2

Sources: National authorities and IMF staff estimates

7. The baseline scenario assumes that fiscal financing needs will be met from external concessional sources in the near term, and non-concessional financing will be avoided until Roghun power purchase agreements are signed and debt is considered sustainable. During 2020-22, fiscal financing needs are expected to be met mostly from concessional loans and grants, provided by

international finance institutions, export credit agencies, and traditional bilateral partners. Budget support loans and grants are expected to increase in 2020 but projected to remain low in 2021-22. There is uncertainty on the terms of future borrowing. For the purposes of this DSA, staff assume concessional borrowing after 2022 in line with the levels of recent years. The authorities are committed to avoid non-concessional borrowing until the Roghun power-purchase agreements are finalized and debt is considered sustainable. Therefore, residual financing needs are expected to be met from non- concessional borrowing starting only after 2022. Staff projections assume no new domestic financing at market determined rates, in line with the recent experience (see paragraph 3) and authorities' financing plans.

8. The realism tools largely suggest that staff forecasts are realistic. Under the baseline, debt accumulation over the projection horizon is smaller than in recent years. The contribution of primary deficits to future debt accumulation is expected to be lower compared to the past few years due to significant consolidation in medium and long-term. Another important reason for the difference is that the contribution of exchange rate depreciation to external and public debt accumulation is lower than in recent years. This is appropriate as the COVID-19 shock is expected to be temporary and the exchange rate adjustment is expected to be smaller. Staff projections of growth, while lower than estimates from recent years, are in line with weaker remittances and external demand going forward.[6]

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TESTS

9. Tajikistan is assessed to have a medium debt-carrying capacity. Tajikistan's debt carrying capacity is currently assessed as medium based on the CI Index from the October 2019 WEO and the World Bank's 2018 CPIA. The 2017 DSA assessed Tajikistan's debt carrying capacity as weak. The upgrade from weak to medium debt-carrying capacity is mainly driven by a shift from CPIA to a composite indicator, which also incorporates other factors such as the import coverage of reserves and remittances.

Debt Carrying Capacity	Medium

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Medium	Medium 2.90	Medium 2.83	Medium 2.90

10. Stress tests for PPPs' agreements, potential size of a rescue of the financial sector, and a commodity price shock are set at default levels. Stress test for PPPs' demand and guarantees is set at a default 1.73 percent of GDP. Stress test for the banking sector is set at default 5 percent of GDP. Default tailored tests for commodity prices are also applied since non-fuel commodity exports constitute an important part of Tajikistan's exports.

11. Tailored contingent liability stress test is designed to incorporate contingent liabilities from potential non-guaranteed debt of SOEs. The debt coverage for Tajikistan excludes non-government guaranteed debt of non-financial public corporations (NFPC) under the baseline given uncertainties on the

[6] There are weaknesses in national accounts statistics.

nature of the debt and lack of full financial information on SOEs. To illustrate the effects of contingent liabilities associated with large SOE debt that might have significant implications for debt sustainability, the size of shock is set at 10 percent of GDP. The shock reflects: (i) 6.5 percent of GDP based on available information on Barki Tojik arrears[7], which could be transferred onto the government's balance sheet; and (ii) 3.5 percent of GDP based on the loan agreements of Tajiktransgaz. Planned borrowings related to the modernization of the TALCO aluminum plant could pose additional contingent liability risks (6½ percent of GDP). However, it is not clear at this stage if the contract to finance the aluminum plant will materialize and what the contract would look like, including whether it would be considered debt, so it is not included in this DSA.

1	The country's coverage of public debt	The central government plus social security and extra budgetary funds, central bank, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	10	Great uncertainty about the true size of liabilities and weak financial position and performance of SOEs
4	PPP	35 percent of PPP stock	1.73	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5	
	Total (2+3+4+5) (in percent of GDP)	16.7		

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

DEBT SUSTAINABILITY

B. External Debt Sustainability Analysis

12. Under the baseline scenario, external debt indicators improve significantly in comparison to 2019 DSA. Under the 2019 DSA, three indicators breached and remained above thresholds, rising steadily during the projection horizon. Under the RCF baseline scenario debt is stabilized in the medium-term. Initial breaches in two external debt indicators in 2020 are partly due to financing needs associated with authorities' COVID-19 response and fall below or near the threshold in the medium term. Both indicators fall below 2020 levels by the end of projection horizon in 2030.

13. Two external debt indicators start above respective thresholds in the first year of projections but fall below or near the threshold in the medium term (Figure 1). Baseline breaches involve one stock/solvency (PV debt-to-export ratio) and one flow/liquidity (debt-service-to exports ratio and debt-service-to revenue ratio) indicator. The flow indicator (debt service-to-exports ratio) is on downward path in and falls below threshold in the medium term. The temporary pick up in 2025 is due to Eurobond principal repayment and it stabilizes below the threshold by the end of the projection horizon.[8] The other flow indicator, debt service-to-revenue ratio, remains below threshold under baseline. Both solvency indicators are stable throughout the projection horizon. The initial breach in debt-to-exports ratio is mild and partly

[7] Barki-Tojik is a state-owned energy company in Tajikistan.

[8] A successful refinancing of the Eurobond would smooth the repayment profile and the profile of the external debt service indicators.

due to new borrowing to finance COVID-19 spending. The ratio declines near the respective threshold throughout the projection horizon.

14. Under the stress scenarios, all external debt indicators breach respective thresholds. Breaches in debt-to-exports ratio and debt-service-to exports are significant and point to debt vulnerabilities. Shocks to exports are the most extreme and impactful for these indicators. Under a shock to exports, the PV of debt-to-exports ratio reaches about 346 percent (versus 180 percent threshold), while the PV of debt service-to-exports ratio reaches 27 percent (versus 15 percent threshold) by the end of horizon. The contingent liability shock also causes a deterioration in external debt sustainability. The trajectory of the PV of external public debt-to-GDP ratio moves upwards by close to 11 percentage points from the baseline. This suggests the need for the government to improve debt recording and management practices (especially for SOEs) and rebuild fiscal buffers to address the rising contingent liabilities.

15. The market-financing risks are moderate in the short-term. Recent tightening of global financial conditions due to COVID19 pandemic has pushed the spread on Tajikistan's sovereign bond (1500bps) further beyond the benchmark (570bps) under the market module. Although gross financing needs in 2020 rise sharply due to response to COVID-19 pandemic it does not breach the 14 percent threshold. Authorities commitment to avoid non-concessional borrowing and commitment to fiscal consolidation should allow the market financing risk to be moderate or low in the medium term.

C. Public Sector Debt Sustainability Analysis

16. Under the baseline, overall public debt improves significantly and points to lower risk in comparison to the 2019 DSA. The public debt burden indicator (PV total debt-to-GDP) ratio stabilizes and remains below the 55 percent benchmark throughout the projection horizon.

17. The standardized sensitivity analysis shows lower risks as the indicator remains below threshold under stress scenario. Shock to combined contingent liabilities leads to the highest public debt figures. The shock causes a 12 percent deterioration in comparison to baseline debt ratio by 2030. This highlights the need for strengthened oversight of SOE sector and streamlined borrowing policies at a time when the government is already financing a large infrastructure project.

CONCLUSION AND RISK RATING

18. The debt sustainability analysis under the new LIC DSF framework suggests that Tajikistan's risk of external and overall public debt distress is high. These results are similar to the 2019 DSA findings, but debt stabilizes under the baseline.

19. Tajikistan's risk of external debt distress remains high. Two external debt-burden (debt service-to-exports ratio and PV of debt-to-exports ratio) indicators breach respective thresholds under the baseline. The Debt service-to-exports indicator stabilizes after the Eurobond repayment is completed and falls below the threshold by 2028. The PV of debt-to-exports ratio is on a downward path and but stays slightly above the threshold by the end of the horizon. In comparison to 2019 DSA, all debt burden ratios are stabilized during the projection horizon. External debt is most vulnerable to exports shocks and contingent liabilities.

Baseline scenario and standardized stress tests indicate the importance of containing contingent liabilities and broadening the export base.

20. Overall risk of public debt distress is high under the baseline due to breaches in external debt indicators. A contingent liability shock has the largest impact on public debt sustainability.

21. Under the baseline Tajikistan's public debt becomes sustainable owing to medium and long-term fiscal consolidation. External debt indicators that breach respective thresholds under the baseline fall below or near the threshold. All stock and flow indicators are on a stable trajectory during the projection horizon. It is also worth noting that while external debt risks are high, total public debt levels do not breach benchmarks in both the baseline and stress tests. Debt is assessed to be sustainable based on the authorities' commitment to fiscal consolidation measures of 2 percent of GDP in 2021-22 and to avoid non-concessional borrowing. The planned fiscal adjustment is a mix of revenue and expenditure measures. While the detailed revenue and expenditure measures will be specified in the coming months, the authorities plan to increase in tax revenues by 0.5 percent of GDP by phasing out tax incentives and broadening the tax base. They also plan to reduce capital expenditures by 1.5 percent of GDP in 2020, reflecting better prioritization of spending and improvements to the efficiency of public investment projects. In the case fiscal adjustment falls short or the authorities resort to non-concessional borrowing, the debt path may deteriorate, putting debt sustainability under pressure. A more severe or prolonged COVID-19 shock could heighten debt vulnerabilities. On the other hand, greater-than-expected progress with economic diversification or higher energy and non-energy exports would improve debt sustainability over the longer term.

22. Other measures should also be taken to reduce debt vulnerabilities. Diversifying exports and containing contingent liabilities will reduce the vulnerabilities of public debt to shocks. Improving debt management practices, including by smoothing the repayment profile could help address large breaches in the debt service-to exports ratio in the medium term. Further upgrading the debt recording and reporting practices, and enhancing the linkages between the medium-term debt management strategy and the government's borrowing plans would further help to contain debt vulnerabilities.

AUTHORITIES' VIEWS

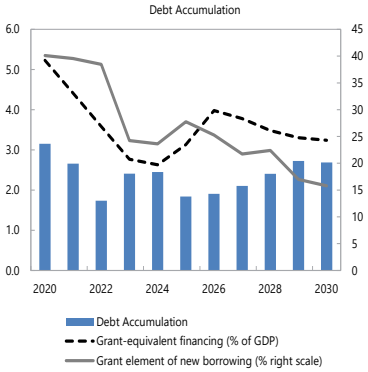
23. The authorities broadly agreed with overall assessment. They concurred with staff that debt vulnerabilities need to be better managed, including through a medium-term fiscal adjustment and seeking concessional borrowing to meet financing needs. They agreed that non-concessional borrowing would weaken debt sustainability and in case of such borrowing they would use it to refinance more expensive credits to improve their debt profile.

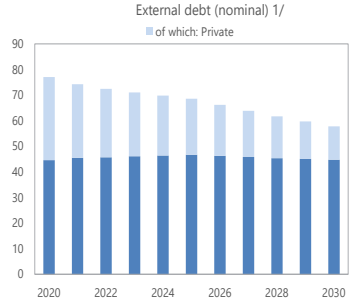
Table 1. Tajikistan: External Debt Sustainability Framework, Baseline Scenario, 2019-40

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 8/	
	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
External debt (nominal) 1/	**68.5**	**77.0**	**74.2**	**72.4**	**71.0**	**69.8**	**68.6**	**57.8**	**47.5**	**59.4**	**67.5**
of which: public and publicly guaranteed (PPG)	**36.3**	**44.6**	**45.5**	**45.7**	**46.1**	**46.4**	**46.7**	**44.8**	**44.1**	**32.3**	**45.7**
Change in external debt	-2.0	8.5	-2.8	-1.8	-1.4	-1.2	-1.2	-1.9	-0.4		
Identified net debt-creating flows	**-5.2**	**8.0**	**-3.3**	**-0.8**	**-0.4**	**-0.5**	**-0.5**	**-0.7**	**-0.3**	**0.5**	**-0.1**
Non-interest current account deficit	**1.5**	**5.6**	**2.5**	**2.5**	**2.3**	**2.2**	**2.0**	**1.4**	**1.1**	**4.5**	**2.3**
Deficit in balance of goods and services	26.7	23.9	24.8	24.4	24.2	23.9	23.4	22.9	22.9	**36.6**	**23.6**
Exports	15.3	14.8	16.0	16.7	17.0	17.4	17.8	18.3	18.3		
Imports	42.0	38.6	40.8	41.0	41.2	41.3	41.2	41.2	41.2		
Net current transfers (negative = inflow)	-8.0	-4.7	-6.4	-6.3	-6.4	-6.3	-6.3	-6.3	-6.3	**-9.1**	**-6.1**
of which: official	-1.4	-0.3	-0.3	-0.5	-0.5	-0.5	-0.5	-0.5	0.0		
Other current account flows (negative = net inflow)	-17.2	-13.6	-15.9	-15.5	-15.5	-15.4	-15.2	-15.2	-15.6	**-23.0**	**-15.2**
Net FDI (negative = inflow)	**-2.3**	**-1.2**	**-2.5**	**-2.3**	**-2.0**	**-2.1**	**-2.0**	**-2.0**	**-2.0**	**-2.9**	**-2.0**
Endogenous debt dynamics 2/	**-4.3**	**3.6**	**-3.3**	**-1.1**	**-0.7**	**-0.6**	**-0.5**	**-0.1**	**0.6**		
Contribution from nominal interest rate	0.8	2.2	2.1	2.1	2.1	2.1	2.1	2.1	2.4		
Contribution from real GDP growth	-4.9	1.4	-5.5	-3.2	-2.8	-2.7	-2.6	-2.2	-1.8		
Contribution from price and exchange rate changes	-0.2	...	...	...	...	...	...	...	...		
Residual 3/	**3.2**	**0.5**	**0.5**	**-1.0**	**-1.0**	**-0.8**	**-0.7**	**-1.2**	**-0.1**	**2.8**	**-0.9**
of which: exceptional financing	0.0	-0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**28.3**	**32.5**	**33.2**	**33.1**	**33.8**	**34.5**	**34.5**	**34.9**	**36.6**		
PV of PPG external debt-to-exports ratio	**184.4**	**220.1**	**207.5**	**198.9**	**198.4**	**198.1**	**194.1**	**190.3**	**199.9**		
PPG debt service-to-exports ratio	**13.1**	**18.9**	**16.7**	**15.5**	**13.0**	**12.4**	**22.1**	**14.1**	**21.6**		
PPG debt service-to-revenue ratio	**8.0**	**11.9**	**11.0**	**10.5**	**8.9**	**8.6**	**15.6**	**10.7**	**16.2**		
Gross external financing need (Million of U.S. dollars)	1123.4	2078.5	1621.4	1791.0	1929.0	1830.9	1796.1	1458.5	1125.9		
Key macroeconomic assumptions											
Real GDP growth (in percent)	7.5	-2.0	7.5	4.5	4.0	4.0	4.0	4.0	4.0	**7.0**	**3.8**
GDP deflator in US dollar terms (change in percent)	0.3	-1.4	-1.5	0.9	1.0	1.1	1.4	2.6	2.6	**-1.4**	**1.3**
Effective interest rate (percent) 4/	1.2	3.1	3.0	3.0	3.0	3.2	3.2	3.8	5.4	**2.0**	**3.2**
Growth of exports of G&S (US dollar terms, in percent)	11.4	-6.9	14.8	9.8	7.6	7.4	7.5	7.1	6.7	**9.4**	**7.0**
Growth of imports of G&S (US dollar terms, in percent)	5.9	-11.2	11.7	6.2	5.5	5.3	5.2	6.7	6.7	**3.2**	**5.1**
Grant element of new public sector borrowing (in percent)	...	40.1	39.5	38.5	24.2	23.6	27.7	15.8	16.9	**...**	**26.9**
Government revenues (excluding grants, in percent of GDP)	25.2	23.5	24.2	24.7	25.0	25.1	25.2	24.3	24.3	**25.1**	**24.5**
Aid flows (in Million of US dollars) 5/	181.2	522.1	453.9	350.8	209.1	208.7	338.2	524.8	1018.7		
Grant-equivalent financing (in percent of GDP) 6/	...	5.2	4.4	3.6	2.8	2.6	3.1	3.2	3.4	**...**	**3.6**
Grant-equivalent financing (in percent of external financing) 6/	...	53.2	54.8	57.4	48.6	47.8	44.8	47.2	44.2	**...**	**49.9**
Nominal GDP (Million of US dollars)	8,110	7,836	8,301	8,757	9,200	9,672	10,199	14,051	26,777		
Nominal dollar GDP growth	7.8	-3.4	5.9	5.5	5.1	5.1	5.4	6.7	6.7	**5.5**	**5.2**
Memorandum items:											
PV of external debt 7/	60.5	65.0	62.0	59.8	58.7	57.9	56.3	47.9	39.9		
In percent of exports	394.9	439.6	387.1	359.2	344.4	332.2	317.4	261.2	218.4		
Total external debt service-to-exports ratio	95.8	149.7	122.0	121.2	121.3	108.0	99.3	59.8	28.3		
PV of PPG external debt (in Million of US dollars)	2292.5	2548.0	2756.3	2900.2	3111.1	3336.4	3514.2	4905.6	9790.0		
(PVt-PVt-1)/GDPt-1 (in percent)		3.2	2.7	1.7	2.4	2.4	1.8	2.7	2.3		
Non-interest current account deficit that stabilizes debt ratio	3.5	-2.9	5.3	4.3	3.7	3.4	3.2	3.3	1.5		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + Ɛα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, Ɛ=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

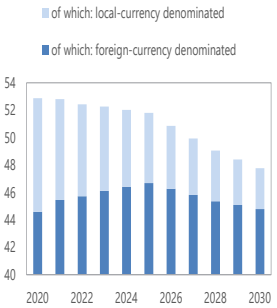
Table 2. Tajikistan: Public Sector Debt Sustainability Framework, Baseline Scenario, 2019-40

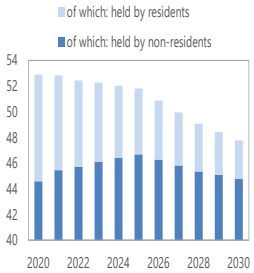
(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 6/	
	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
Public sector debt 1/	**44.6**	**52.9**	**52.8**	**52.4**	**52.3**	**52.0**	**51.8**	**47.8**	**45.0**	**38.1**	**50.9**
of which: external debt	**36.3**	**44.6**	**45.5**	**45.7**	**46.1**	**46.4**	**46.7**	**44.8**	**44.1**	**32.3**	**45.7**
Change in public sector debt	-3.3	8.3	-0.1	-0.4	-0.2	-0.2	-0.2	-0.6	-0.3		
Identified debt-creating flows	**-3.0**	**7.5**	**-0.8**	**-1.2**	**-1.1**	**-1.1**	**-1.1**	**-0.6**	**-0.3**	**0.1**	**-0.1**
Primary deficit	**1.2**	**6.6**	**3.5**	**1.8**	**1.7**	**1.6**	**1.6**	**1.8**	**1.5**	**2.1**	**2.4**
Revenue and grants	27.4	25.6	26.3	26.6	26.8	26.9	26.9	26.9	26.9	**27.5**	**26.7**
of which: grants	2.2	2.1	2.0	1.9	1.8	1.7	1.7	2.6	2.6		
Primary (noninterest) expenditure	28.6	32.2	29.7	28.4	28.5	28.5	28.4	28.7	28.4	**29.5**	**29.1**
Automatic debt dynamics	**-3.7**	**1.0**	**-4.3**	**-3.0**	**-2.7**	**-2.7**	**-2.7**	**-2.4**	**-1.8**		
Contribution from interest rate/growth differential	-2.4	1.0	-4.3	-3.0	-2.7	-2.7	-2.7	-2.4	-1.8		
of which: contribution from average real interest rate	0.9	0.1	-0.6	-0.7	-0.7	-0.7	-0.7	-0.6	-0.1		
of which: contribution from real GDP growth	-3.3	0.9	-3.7	-2.3	-2.0	-2.0	-2.0	-1.9	-1.7		
Contribution from real exchange rate depreciation	-1.3	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**-0.6**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**-0.1**	**0.0**
Privatization receipts (negative)	-0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-0.3**	**0.8**	**0.7**	**0.8**	**0.9**	**0.9**	**0.8**	**-0.1**	**0.0**	**0.7**	**0.4**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**37.1**	**42.2**	**41.2**	**40.5**	**40.6**	**40.8**	**40.4**	**38.6**	**38.2**		
PV of public debt-to-revenue and grants ratio	**135.3**	**164.7**	**156.8**	**152.2**	**151.7**	**151.7**	**150.2**	**143.8**	**142.2**		
Debt service-to-revenue and grants ratio 3/	**9.4**	**12.4**	**12.7**	**12.4**	**10.7**	**10.3**	**16.7**	**10.8**	**14.6**		
Gross financing need 4/	3.2	9.7	6.8	5.1	4.5	4.4	6.1	4.7	5.4		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	7.5	-2.0	7.5	4.5	4.0	4.0	4.0	4.0	4.0	**7.0**	**3.8**
Average nominal interest rate on external debt (in percent)	1.3	2.5	2.3	2.0	2.0	2.1	2.1	2.1	3.0	**0.2**	**2.1**
Average real interest rate on domestic debt (in percent)	-2.3	-2.5	-4.4	-4.8	-4.8	-4.9	-5.0	-6.3	-6.5	**-1.5**	**-5.2**
Real exchange rate depreciation (in percent, + indicates depreciation)	-3.4	...	...	...	...	...	...	...	...	**1.8**	**...**
Inflation rate (GDP deflator, in percent)	4.4	5.6	5.2	5.3	5.2	5.4	5.4	6.9	6.9	**6.7**	**6.1**
Growth of real primary spending (deflated by GDP deflator, in percent)	0.1	10.2	-0.8	-0.3	4.4	4.1	3.8	3.5	4.1	**7.8**	**3.9**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	4.6	-1.7	3.5	2.2	1.8	1.8	1.8	2.5	1.8	**2.0**	**2.1**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government plus social security and extra budgetary funds, central bank, government-guaranteed debt . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

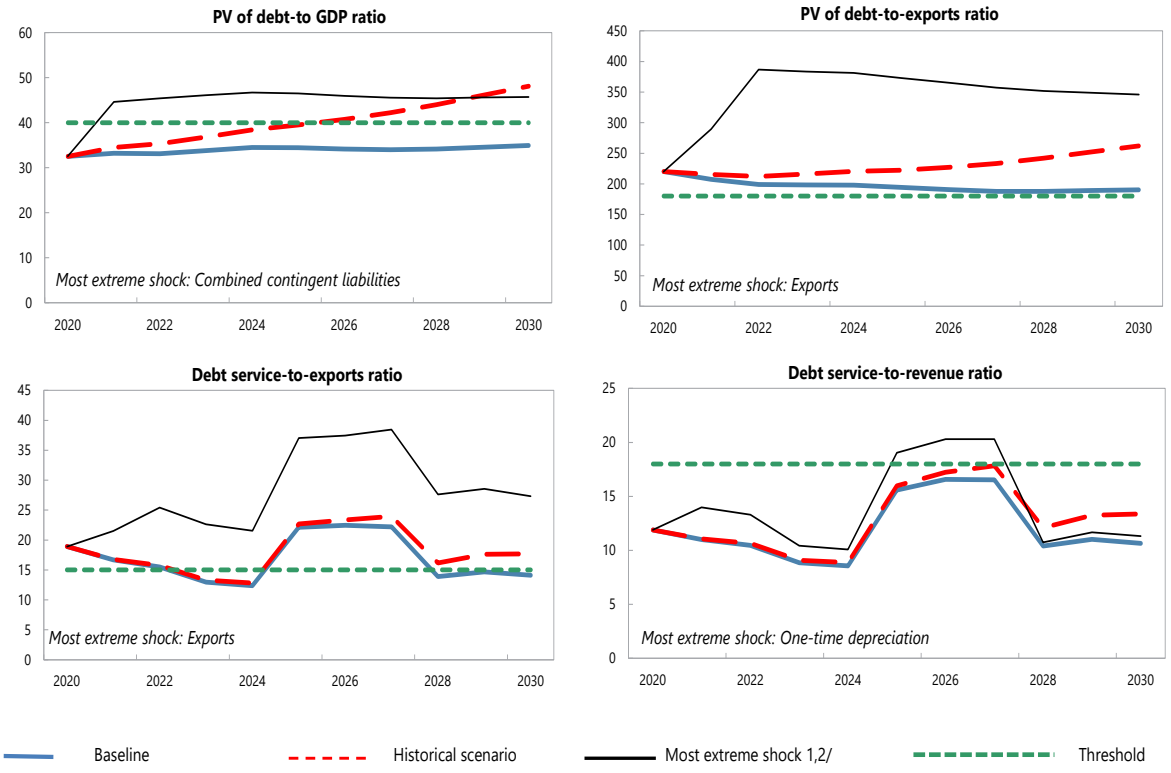
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Tajikistan: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2020–30



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	n.a.	n.a.
Commodity price	No	No
Market financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.3%	2.3%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	20	20
Avg. grace period	5	5

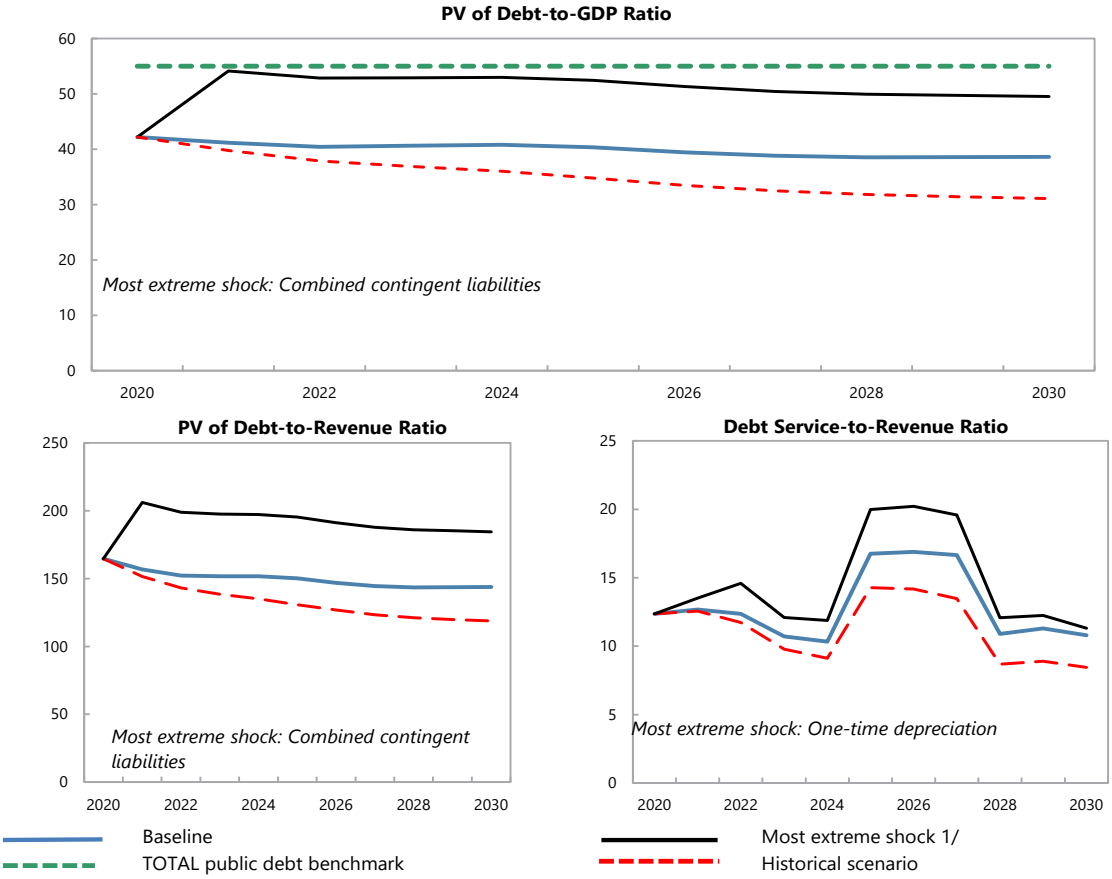
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Tajikistan: Indicators of Public Debt Under Alternative Scenarios, 2020–30



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	91%	91%
Domestic medium and long-term	0%	0%
Domestic short-term	9%	9%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.3%	2.3%
Avg. maturity (incl. grace period)	20	20
Avg. grace period	5	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	0.0%	0.0%
Avg. maturity (incl. grace period)	1	1
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	-3.0%	-3.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Tajikistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2020-30

(In percent)

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of debt-to GDP ratio											
Baseline	33	33	33	34	34	34	34	34	34	35	35
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	33	34	35	37	38	39	**41**	**42**	**44**	**46**	**48**
B. Bound Tests											
B1. Real GDP growth	33	34	33	33	34	34	34	33	34	34	34
B2. Primary balance	33	35	39	40	**41**	**40**	40	40	40	40	40
B3. Exports	33	36	**41**	**42**	**43**	**43**	**42**	**42**	**41**	**41**	**41**
B4. Other flows 3/	33	36	38	39	39	39	39	38	38	38	38
B5. Depreciation	33	**42**	35	36	37	37	37	37	37	39	40
B6. Combination of B1-B5	33	38	37	37	38	38	37	37	37	37	37
C. Tailored Tests											
C1. Combined contingent liabilities	33	**45**	**45**	**46**	**47**	**47**	**46**	**46**	**45**	**46**	**46**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	33	35	36	37	37	36	36	35	34	34	34
C4. Market Financing	33	37	37	38	39	39	38	38	38	39	39
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	**220**	**207**	**199**	**198**	**198**	**194**	**190**	**188**	**188**	**189**	**190**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	**220**	**216**	**212**	**216**	**221**	**222**	**227**	**233**	**242**	**252**	**262**
B. Bound Tests											
B1. Real GDP growth	**220**	**207**	**199**	**198**	**198**	**194**	**190**	**188**	**188**	**189**	**190**
B2. Primary balance	**220**	**220**	**235**	**234**	**233**	**228**	**223**	**219**	**217**	**218**	**218**
B3. Exports	**220**	**289**	**387**	**384**	**381**	**373**	**365**	**357**	**352**	**349**	**346**
B4. Other flows 3/	**220**	**223**	**228**	**227**	**226**	**221**	**217**	**212**	**210**	**209**	**208**
B5. Depreciation	**220**	**207**	166	167	167	164	162	160	162	166	170
B6. Combination of B1-B5	**220**	**262**	**219**	**279**	**278**	**272**	**266**	**261**	**259**	**259**	**259**
C. Tailored Tests											
C1. Combined contingent liabilities	**220**	**279**	**273**	**270**	**268**	**262**	**256**	**252**	**250**	**250**	**249**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**220**	**240**	**234**	**228**	**222**	**211**	**201**	**195**	**190**	**188**	**186**
C4. Market Financing	**220**	**207**	**199**	**198**	**198**	**194**	**191**	**188**	**187**	**189**	**190**
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	19	17	16	13	12	22	22	22	14	15	14
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	19	17	16	13	13	23	23	24	16	18	18
B. Bound Tests											
B1. Real GDP growth	19	17	16	13	12	22	22	22	14	15	14
B2. Primary balance	19	17	16	15	14	24	24	24	17	17	17
B3. Exports	19	22	25	23	22	37	37	38	28	29	27
B4. Other flows 3/	19	17	16	14	13	23	23	24	16	17	16
B5. Depreciation	19	17	16	12	11	21	22	21	11	12	12
B6. Combination of B1-B5	19	19	21	17	16	29	29	30	20	21	20
C. Tailored Tests											
C1. Combined contingent liabilities	19	17	18	15	14	24	24	24	16	16	16
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	19	19	17	15	14	24	24	24	15	16	15
C4. Market Financing	19	17	16	13	13	22	23	24	16	15	14
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	12	11	10	9	9	16	17	17	10	11	11
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	12	11	11	9	9	16	17	18	12	13	13
B. Bound Tests											
B1. Real GDP growth	12	11	10	9	8	15	16	16	10	11	10
B2. Primary balance	12	11	11	10	10	17	18	**18**	12	13	13
B3. Exports	12	11	11	10	10	17	18	**18**	13	14	13
B4. Other flows 3/	12	11	11	9	9	16	17	18	12	13	12
B5. Depreciation	12	14	13	10	10	**19**	**20**	**20**	11	12	11
B6. Combination of B1-B5	12	11	11	9	9	16	17	18	12	12	12
C. Tailored Tests											
C1. Combined contingent liabilities	12	11	12	10	10	17	18	18	12	12	12
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	12	12	11	10	9	17	17	17	11	12	11
C4. Market Financing	12	11	10	9	9	16	17	**18**	12	11	11
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Tajikistan: Sensitivity Analysis for Key Indicators of Public Debt, 2020-30

(In percent)

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of Debt-to-GDP Ratio											
Baseline	42	41	40	41	41	40	39	39	39	39	39
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	42	40	38	37	36	35	33	32	32	31	31
B. Bound Tests											
B1. Real GDP growth	42	42	40	40	40	39	38	37	36	36	36
B2. Primary balance	42	43	47	47	47	46	45	45	44	44	44
B3. Exports	42	44	48	48	48	48	47	46	45	44	44
B4. Other flows 3/	42	44	45	46	46	45	44	43	43	42	42
B5. Depreciation	42	47	44	43	41	39	36	34	33	32	31
B6. Combination of B1-B5	42	41	41	40	39	38	36	34	33	33	32
C. Tailored Tests											
C1. Combined contingent liabilities	42	54	53	53	53	52	51	50	50	50	50
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	42	43	43	44	45	44	43	42	41	41	40
C4. Market Financing	42	41	40	41	41	40	40	39	38	39	39
TOTAL public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	165	157	152	152	152	150	147	145	143	144	144
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	165	152	143	139	135	131	127	123	121	120	119
B. Bound Tests											
B1. Real GDP growth	165	160	150	148	147	144	140	137	135	134	133
B2. Primary balance	165	165	176	175	175	173	169	166	164	163	163
B3. Exports	165	167	180	179	179	177	174	170	167	165	163
B4. Other flows 3/	165	166	171	170	170	168	165	161	159	158	156
B5. Depreciation	165	180	168	161	155	146	136	128	123	119	115
B6. Combination of B1-B5	165	156	154	149	146	141	135	129	125	122	119
C. Tailored Tests											
C1. Combined contingent liabilities	165	206	199	198	197	195	191	188	186	185	184
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	165	171	169	172	172	169	163	156	153	151	149
C4. Market Financing	165	157	152	152	152	150	147	145	143	143	143
Debt Service-to-Revenue Ratio											
Baseline	12	13	12	11	10	17	17	17	11	11	11
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	12	13	12	10	9	14	14	13	9	9	8
B. Bound Tests											
B1. Real GDP growth	12	13	12	11	10	16	17	16	11	11	10
B2. Primary balance	12	13	14	13	12	18	18	18	13	13	13
B3. Exports	12	13	13	12	11	18	18	18	13	14	13
B4. Other flows 3/	12	13	13	11	11	17	17	18	12	13	12
B5. Depreciation	12	14	15	12	12	20	20	20	12	12	11
B6. Combination of B1-B5	12	12	12	12	10	16	16	16	11	11	10
C. Tailored Tests											
C1. Combined contingent liabilities	12	13	19	13	12	18	18	18	12	12	12
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	12	13	13	12	12	18	18	18	12	12	12
C4. Market Financing	12	13	12	11	10	17	17	18	12	11	11

Sources: Country authorities; and staff estimates and projections.

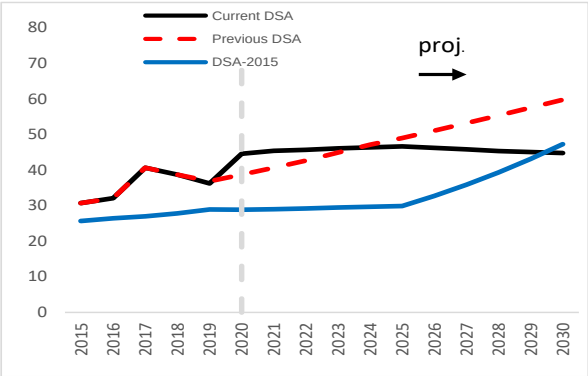
1/ A bold value indicates a breach of the benchmark.

2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

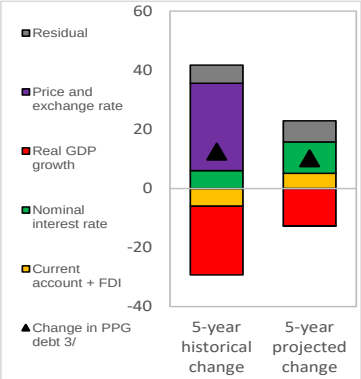
3/ Includes official and private transfers and FDI.

Figure 3. Tajikistan: Drivers of Debt Dynamics – Baseline Scenario External Debt

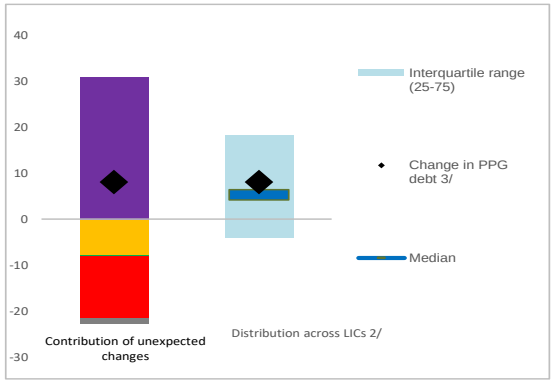
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

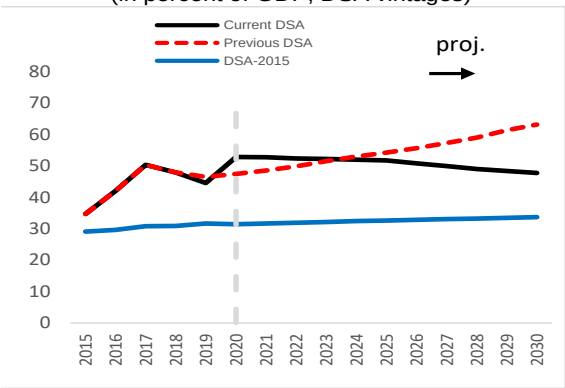


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

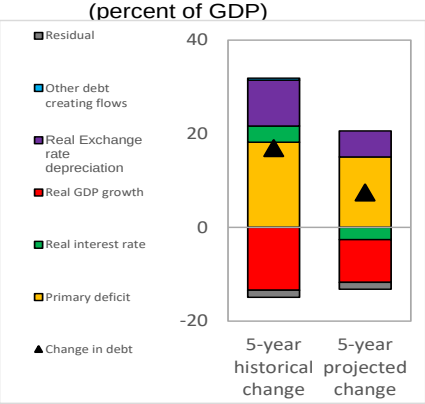


Public debt

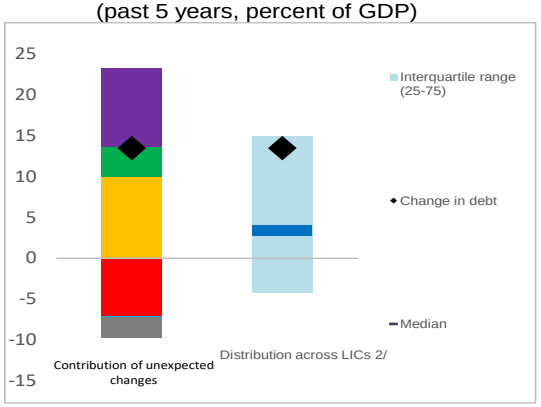
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



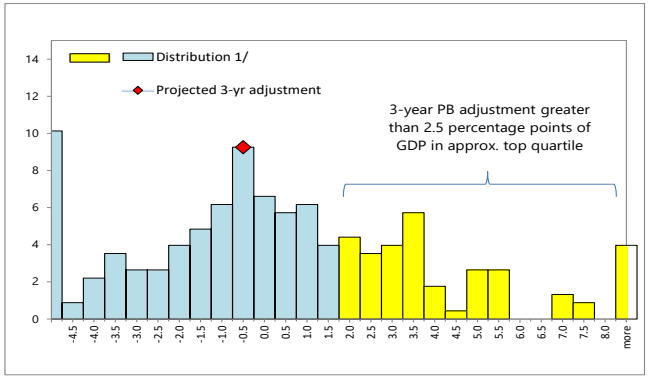
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

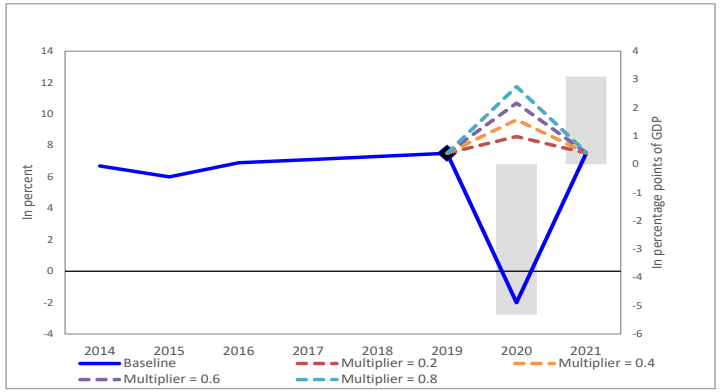
Figure 4. Tajikistan: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



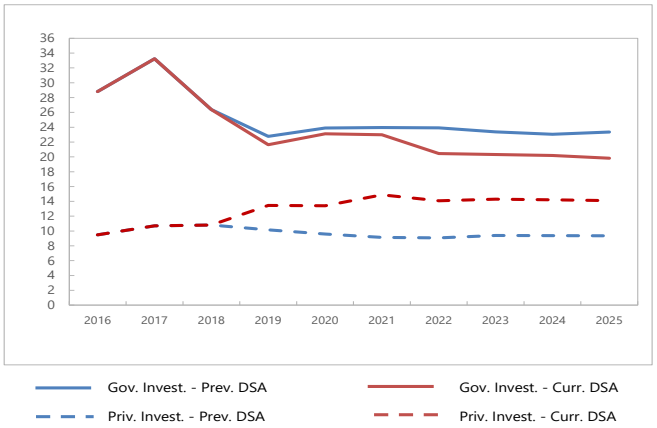
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

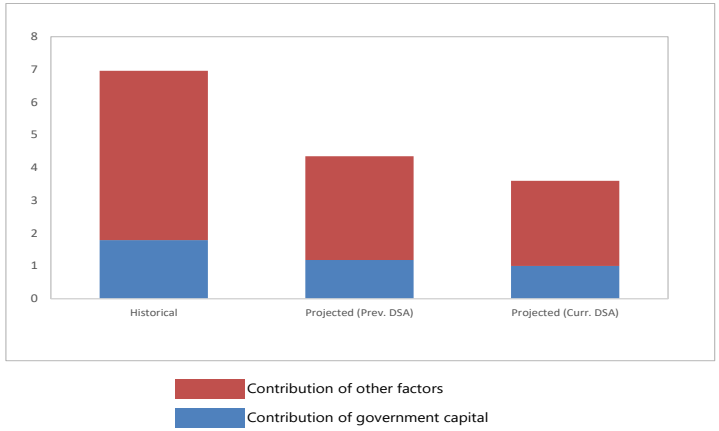
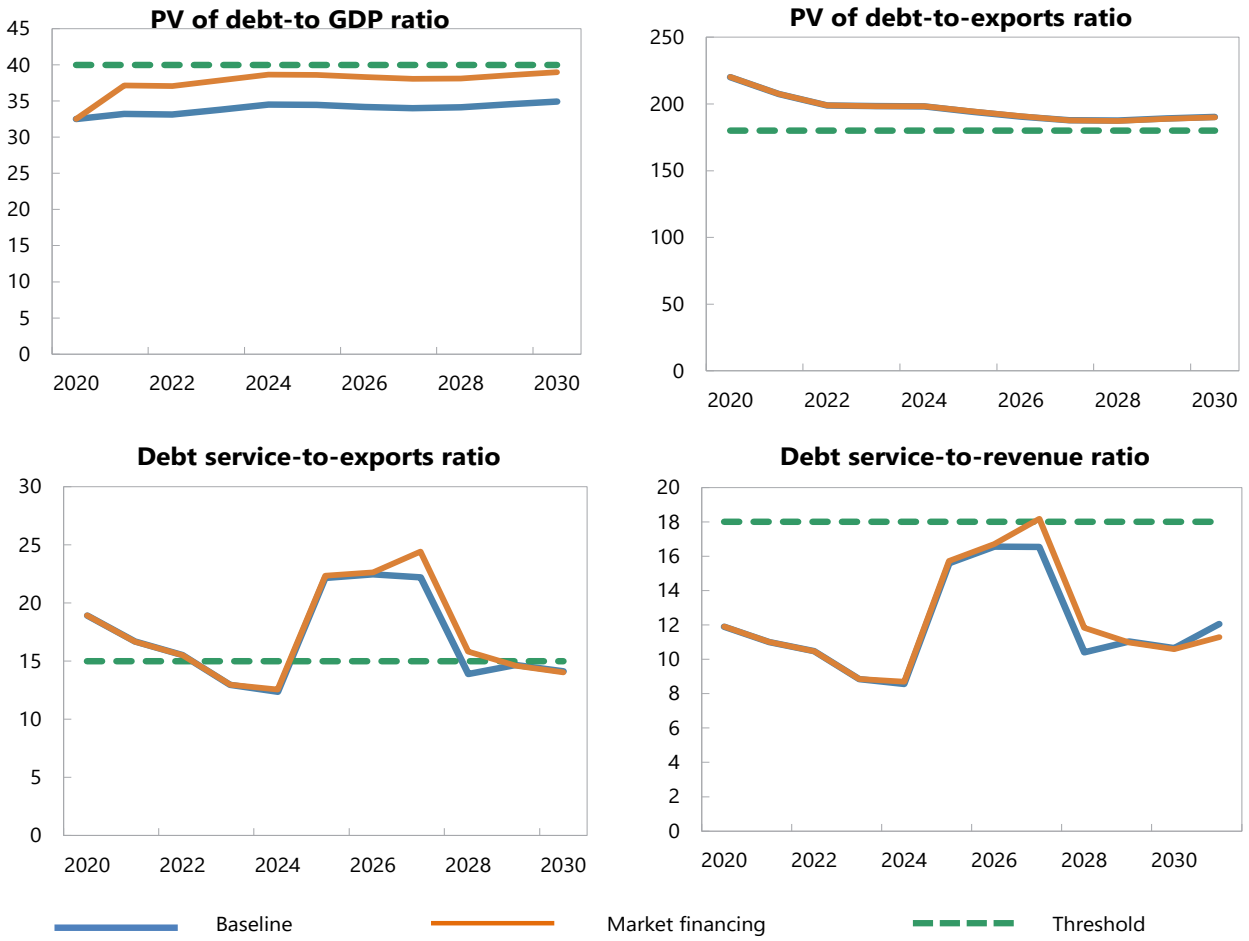


Figure 5. Tajikistan: Market-Financing Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	10	1500
Breach of benchmark	No	Yes
Potential heightened liquidity needs	Moderate	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



Sources: Country authorities; and staff estimates and projections.